

CRISP-DM (Cross-Industry Standard Process for Data Mining) is a framework for data projects that can extend far beyond just data mining. It can be used to complete predictive and general analysis projects as well. It helps to ensure that everything needed further down the business pipeline is completed and understood before building on top of it. Unlike certain workflow frameworks, it allows for falling back to a previous stage to make corrections or change aspects of the project.

CRISP-DM's first phase of six is known as Business Understanding. "This initial phase focuses on understanding the objectives and requirements of the project from a business perspective. Key tasks include: defining business objectives, assessing the current situation, determining (project) goals, producing a project plan" (Chumbar, 2023). This stage is one of the most crucial as it ensures that the project is inline with stakeholder expectations and will provide the business with value. This predictive data project is working with GE's Credit team. Their goal is to train a predictive model (linear regression or neural network) that is able to accurately assess how much credit to extend to its customers based on historical data. The research question being tackled is: Can credit amount be accurately predicted using the historical data at hand?

Being able to accurately disburse credit to customers is a vital to ensuring customers get what they need, and to reducing risk to the business. If GE allocates too little to a customer, then they may not be able to purchase all their resources and will end up taking part if not all of their business elsewhere. If GE allocates too much credit to a customer, it could put them at risk in a couple ways. The customer could be a poor borrower and put the credit at risk, or the customer could not be able to take out too much and have higher payments than they are able to handle. An accurate disbursement not only reduces GE's operational risk, but also ensures customer satisfaction.

The second phase of CRISP-DM is Data Understanding. Like its name implies, this phase is all about gaining an understanding of what data you have to work with and its quality. The data provided by GE can be broken up into a couple of categories. These are: loan characteristics (amount, terms, interest rate, installment), borrower information (annual income, debt to income ratio, whether they own a home), credit history (delinquency in the last 2 years, tax liens, bankruptcies), credit utilization (revolving balance, balance utilization, balance to credit limit, credit limits), and account counts (accounts opened in the past 24 months, total accounts, total bank cards, number of revolving accounts). These features should serve to give a well rounded summary of a customers history with credit and give GE an idea of whether or not they should be lent any money.

This data doesn't contain too many missing values outside of a few exceptions. The largest exception comes from ratio of total current balance to high credit limit. This feature has 624 missing values, or about 12% of total observations. The second highest feature with missing values was num of account currently 120 days past due (updated in past 2 months) with 402 missing values, or about 8%. The next rest of the features had 2% or less missing values or negligible missing values.

After Data Understanding comes the third phase of CRISP-DM, Data Preparation. This phase include a couple main activities, "selecting data, cleaning data, constructing data (feature engineering), integrating data, formatting data" (Chumbar, 2023). These ensure that modeling is conducted on top of a sufficient and organized dataset. In this case, there might be features that are not present, but easily calculated with what is available. Another example is going to be imputing values for missing data.

The fourth phase of CRISP-DM is Modeling. Here is where a lot of the predictive

elements will take place. This phase involves selecting the type of model, the features that will be included in it, and building the training, testing, and evaluation tests needed to properly evaluate the results of the modeling. In this case, GE has asked to either implement a simple linear regression or a small neural network. Other modeling techniques in this business case are going to include feature selection and hyperparameter tuning. The last activity in this case will be implementing evaluation metrics such as F1, recall, precision or ROC-AUC. The model is assessed with these metrics to evaluate quality.

After Modeling, the performance is scrutinized against business objectives. This is a crucial step, to ensure that the final results are in line with what was set out for and adds value to the business. The development process is reviewed as well answering “Was anything overlooked? Were all steps properly executed” (Hotz, 2024). Here, the next steps are determined as well. The project can be pushed to the final phase of development, it can be iterated by returning to an earlier phase, or scrapped for another project.

The final phase of CRISP-DM is deployment. This phase can look very different from organization to organization. According to the CRISP-DM 1.0 Guide, the differences are described as “ [...] the deployment phase can be as simple as generating a report or as complex as implementing a repeatable data mining process across the enterprise” (Chapman, 2000). There are a few completed here will be the same though. Regardless of organizational needs, a report will be generated giving a summary of the project, including results. A project retrospective will also be conducted. This ensure that any mistakes or lessons learned can be carried into the next analytics project.

In this business case, production will look like a complete model that is making accurate predictions on customers credit. The loan amount generated will be used by GE employees to

help accurately dispute the proper loan amount that ensures customer satisfaction and reduced organizational risk. A final report and presentation will also be generated that will explain the technical aspects and results of the project to a non-technical audience.

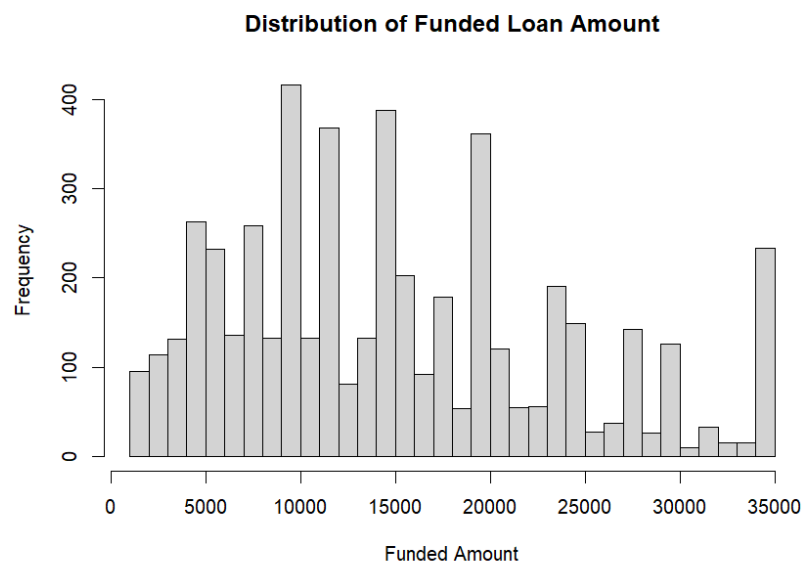
Data Understanding

The source of the data used in this analytics project was given by GE in a CSV format. It contains 5000 observations and 74 numeric features. It contains information on the loans disbursed, credit history, and utilization of the customer, and customer accounts. In order to align with the business's goal of predicting the loan amount to be dispersed to the customer, all the loan terms and characteristics will not be included in the modeling dataset. In fact. Many of the original 74 features will be excluded.

61 features remain after the removal of loan related features. One of the requirements for this project was to use to use a linear regression. A key assumption of linear regression models is that there is a lack of multicollinearity (Geeksforgeeks, 2025). Multicollinearity being when the independent variables are correlated with each other, or put in other terms, they contain redundant information. 6 features of the remaining 61 were calculated to be over 85% collinear and removed from the dataset in order to address the assumptions.

55 features remain, however some of them have almost no correlation to the target variable when calculated. As such, they will not provide much of any information to the model, and possibly serve to confuse it. The dataset was filtered down again based on having a correlation with the target variable greater than 5%. The final dataset contains 34 features and the initial 5000 observations. While containing half the number of features, the removed features either break the linear regression model, or serve to make the model needlessly more complex while providing no useful information.

The second part of the data understanding phase is a statistical analysis of the target and the independent variables. The target variable is the dollar amount of the loan. This feature has a minimum loan amount of 1,000 and a maximum of 35,000. The mean is 15,238 and the median is 14,150, giving it a slight right skew. A histogram was created as well to visually convey the distribution of the target variable, and can be seen below.

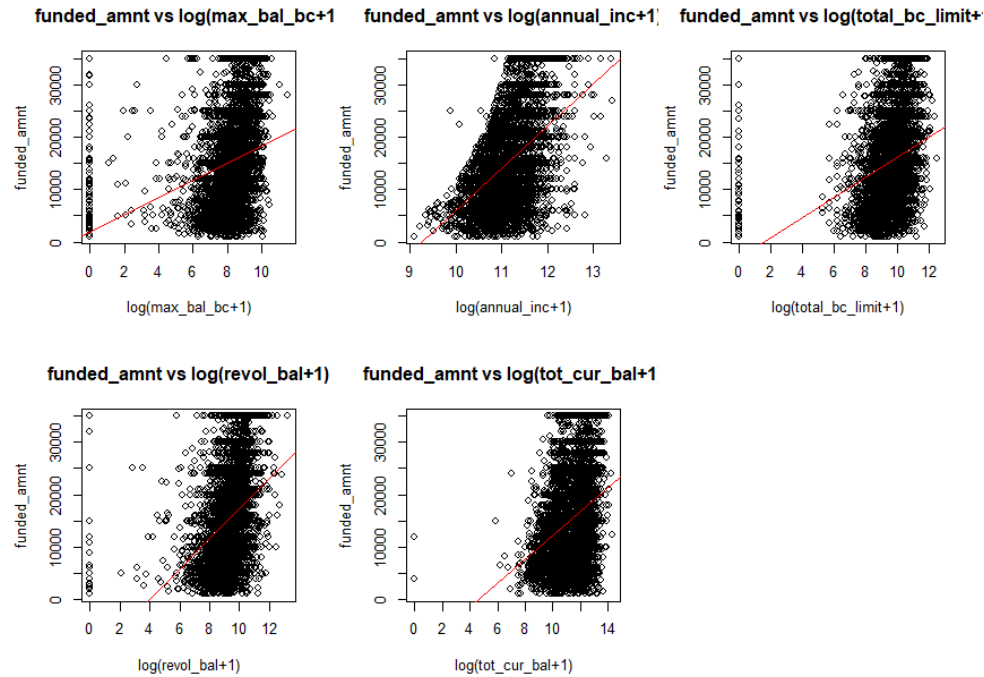


The large spikes are loans that fall on round numbers like 5,000, 10,000, 15,000 and 20,000 for example.

In order to better understand the independent variables, the top 5 most correlated features to the target were examined. Summary statistics were also calculated including the min, max, quartiles, and mean and median. The results of those calculations can be found in the table.

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
max_bal_bc	0	2479	4564.5	5981.067	7865.5	100973
annual_inc	9000	50000	70000	78993.61	98000	675000
total_bc_limit	0	8400	16700	23433.48	30588.75	281300
revol_bal	0	6477.75	12279	17719.49	21494.5	566420
tot_cur_bal	0	34974.25	96319	151756.4	222605	1662218

Scatterplots were also generated to observe the relationship between the independent variables and the target.



All the top features appear to be positively correlated to the target variable. All 5 plots show an upward and to the right trend. The red trend line helps to better identify the pattern in the dense data, though it seems to be off by a little in some of the plots. A logarithmic transformation was applied to all the independent variables in order to normalize the skew for better correlation to the target.

Data Preparation

The first step in data preparation was to gain an understanding of how the data was typed and if there were any errors or missing values. This dataset did in fact contain both errors and missing values. The one error was a simple entry mistake. In one numerical feature, `revol_util_pct`, 23 “-“ entries caused the feature to be interpreted as character data. Rectifying

this was as simple as changing the “-“ to NA and fixing the NAs with the rest of the missing values later.

5 other columns in the final dataset also contained missing values. There are a couple of strategies for imputation. A simple solution like using the median value for that feature, which is robust to outliers. Second is a more complex solution like a KNN model that predicts the missing value based on similar observations. This is more accurate and variable, however, requires training, and evaluation. A KNN would also have to be trained and evaluated for each feature that contains missing values. For the few features I had, with the few missing values, I ended up using median imputation.

Beyond imputations and minor type formatting, the data used in this project didn't need much preparation. No merging or integration was required as all information was contained in one single CSV document. It is now ready for modeling.

After the data preparation phase, the data is ready for modeling. Again, a linear regression model is being selected at the request of GE. The first step in this modeling process was to finalize the data for use in the modeling part of this project. This includes splitting the target from the predictors, and a final feature selection based on the bidirectional stepwise feature selection algorithm.

The data preparation phase identified 33 features that were potentially useful in a model. 33 features could still contain combinations of variables that introduce noise, rather than explaining more of the context of the data. This algorithm works by starting at a null model (one that explains the target with no features) and works its way up to a full model (one that explains the target by all 33 features). At each step in the algorithm, it works through every variable and

adds them to the model and tests the errors. At the end of each step, the variable that had the lowest errors and didn't increase the complexity a lot is added to the base model. This repeats again and again, adding one variable at a time and picking the best variable. Once at the full model, the reverse takes place, removing one variable at a time. This results in a model that balances complexity and has the lowest error across many different combinations for variables.

The greedy nature of this algorithm, picking the best possible choice at all steps instead of exploring all possibilities, means that this algorithm chooses a local best algorithm. There still may be a better model but finding it out of all the hundreds of combinations would take a ton of time and could come at the cost of increasing the modeling complexity. A higher model complexity could in turn decrease the generalization of the model, or its performance on data it has not seen before (V, 2024).

In this datasets case, of the 33 features, the algorithm narrowed down to 17 features that were deemed valuable enough to include and did not increase the model complexity too much. They can all be found below, with their respective directional influence, and p-value.

Feature	Direction	P-value
Max_bal_bc	Positive	0.002833
Annual_inc	Positive	2e-16
Num_op_rev_tl	Positive	0.0000279
Open_acc_6m	Negative	0.0000000246
Total_cu_tl	Positive	0.013249
Delinq_2yrs	Negative	0.000422
Revol_util_pct	Positive	4.61e-15

Total_bc_limit	Positive	0.249878
Total_bal_il	Negative	0.058628
Bc_util	Negative	0.122632
Mort_acc	Negative	0.553782
Pct_tl_nvr_dlq	Positive	0.001612
Il_util	Negative	0.962793
Bc_open_to_buy	Positive	3.38e-13
Avg_cur_bal	Negative	0.678279
Tot_cur_bal	Positive	0.418429
Num_sats	Negative	0.580962

Table 1: All 17 variables used in the final version of the linear model

A negative value in this table indicates whether the feature increases or decreases the amount of the loan. The p-value measures how likely the results are due to random chance. A low p-value, like annual_inc, means that the variable is likely to have a relationship with the outcome. A high p-value, like il_util, means that the variables effect is likely just noise. 9 variables out of these 17 were labeled as significant. They were annual_inc, open_acc_6m, revol_util_pct, bc_open_to_buy, num_op_rev_tl, delinq_2yrs, pct_tl_nvr_dlq, max_bal_bc, and total_cu_tl. These variables are contributing the most to the prediction.

During the model process that has been laid out in this analytic plan, only one data quality concern arose. As was discovered in the visualization of the data correlation, applying a logarithmic scale to the predictors made a linear relationship between the target much more prominent. As such, the same logarithmic scale was applied to the data before modeling as well, both the predictors and target. Not only did this improve the errors made by the model, but also

the r^2 value (how well the predictors explain the target). The final shape of the data ended up being 17 features by 5,000 observations. The predictions are exponentiated to bring them back to dollar amounts for use by GE's employees.

This model was tested by making predictions against the training data and comparing it against the given values. This is the training accuracy, but the test accuracy is also calculated. This is calculated by making predictions on the 100 observations given by GE, that the model has never seen before. This second accuracy will indicate how well the model is at being generalized or making accurate predictions on data it has never seen before.

For the evaluation phase of this project, the models are evaluated in a couple of different ways. Root mean squared error (RSME) is used to measure the root error. The root mean squared error is used over the mean squared error because the root error penalized large errors more. This makes sense in this business case as large errors mean giving significantly more money than should have been allotted. Mean absolute percent error is the same as the average error but described in terms of a percentage.

Metric	Train	Test	Notes
RSME	7487.513	7722.057	Off by ~\$7,700 on average
MAPE	57.501	54.809	~55-57% average percentage error
R2	0.297	0.233	Explains ~23-30% of variance

Table 2: *Train vs Test Evaluation Comparison*

The statistics in Table 2 show the resulting metrics of the final model when predicting on both on the train and test datasets. The RSME for the model shows that the model is off its predictions by \$7,700 dollars from the actual loan value. Between the train and test, the model is

off by only ~\$235, which indicates that the model is not overfitting to the train data. The MAPE contextualizes the dollar amount a bit more, telling us that the model is off by 55% the actual loan amount. The R2 value is the most telling. It highlights the largest issue with this model; the data only describes ~26% of the variance. 74% of the data goes unexplained with the current model.

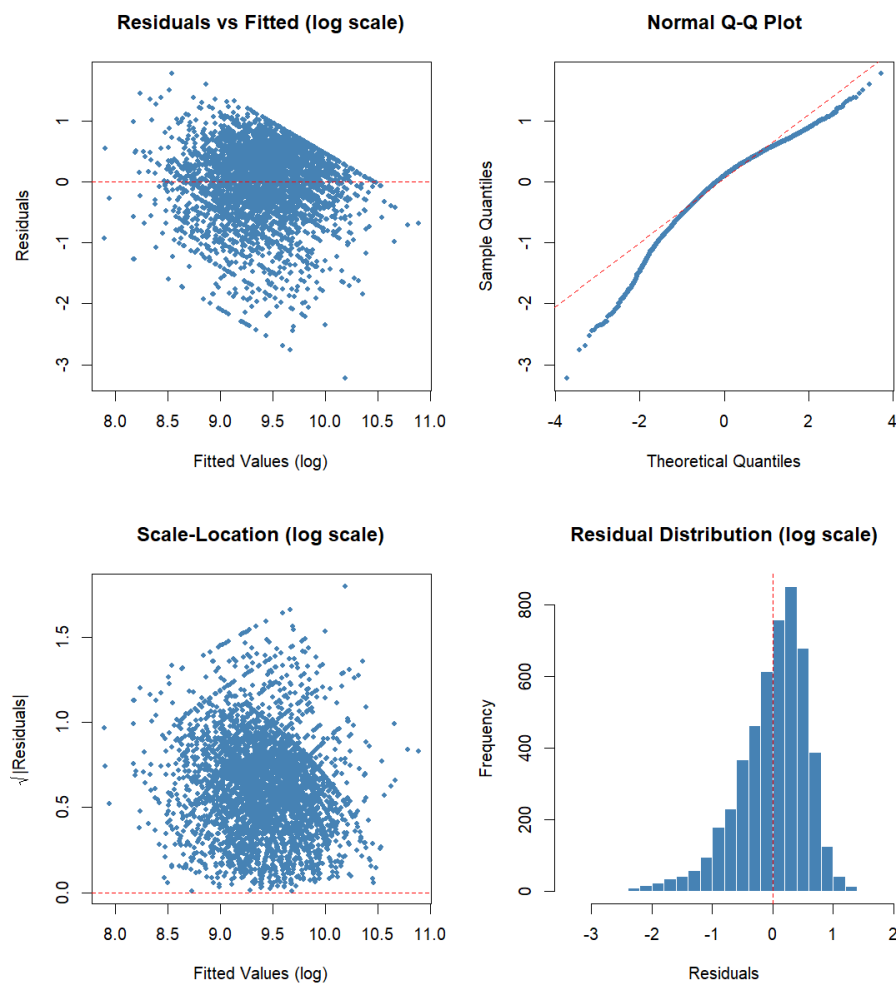


Image 1: *Plots Explaining Model Residuals*

The image above shows a series of plots that help to further explain the residuals (the difference between the predicted value and the actual value). The top left plot shows the residuals

plotted against the true values. The ideal plot here looks random with no discernible pattern in the data. This would mean that the errors are due to noise in the model and not a systemic problem. It would indicate that the linear model fit well to the data. Instead the data seems to slope. There are no values that fall above 0 and above ~10. This could indicate an differing levels of error depending on the loan amount.

The top right plot shows a Q-Q plot, which stands for quantile-quantile. This is a tool that is used to diagnose the distribution of the errors against a perfect distribution. In an ideal linear regression, the plotted points would fall directly on the line. In this case we see the lower section fall quite below the line, the middle section is distributed almost perfectly, and the upper section is slightly below the line. This again reinforces that the model predicts some loan value very well, and struggles with others.

The bottom left plot shows very similar information to the top left plot, but slightly different. This plot shows the square root of the absolute residuals, plotted against the true values. Again, an ideal plot would show randomness without clustering. A bright horizontal spread should be seen indicating that the errors are consistent across the range. This plot shows large and random differences between the residuals with missing parts.

The final plot is a distribution of the residuals. An ideal distribution should show residuals equally likely to be negative or positive. The histogram should follow normal distribution. The plot of the model's residual shows a left-hand skew, again reinforcing that the model struggles with some loan values. This would suggest that the model is not fulfilling the normality assumption of linear regressions (Geeksforgeeks, 2025).

Due to the cyclical nature of modeling and evaluating, it is important to save the plots and statistics regarding this version of the model to compare against. That way you can directly know whether the changes to the model improved it, made it worse, or made no difference at all. It is equally as important to employ good accountability and reproducibility practices like documenting results, keeping code, and setting randomization seeds. This ensures that anyone can reproduce the results explained in this paper.

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